

Experiment 2: Create and Manage an S3 Bucket

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Objective

- Create an S3 bucket and upload, manage, and retrieve files

Introduction

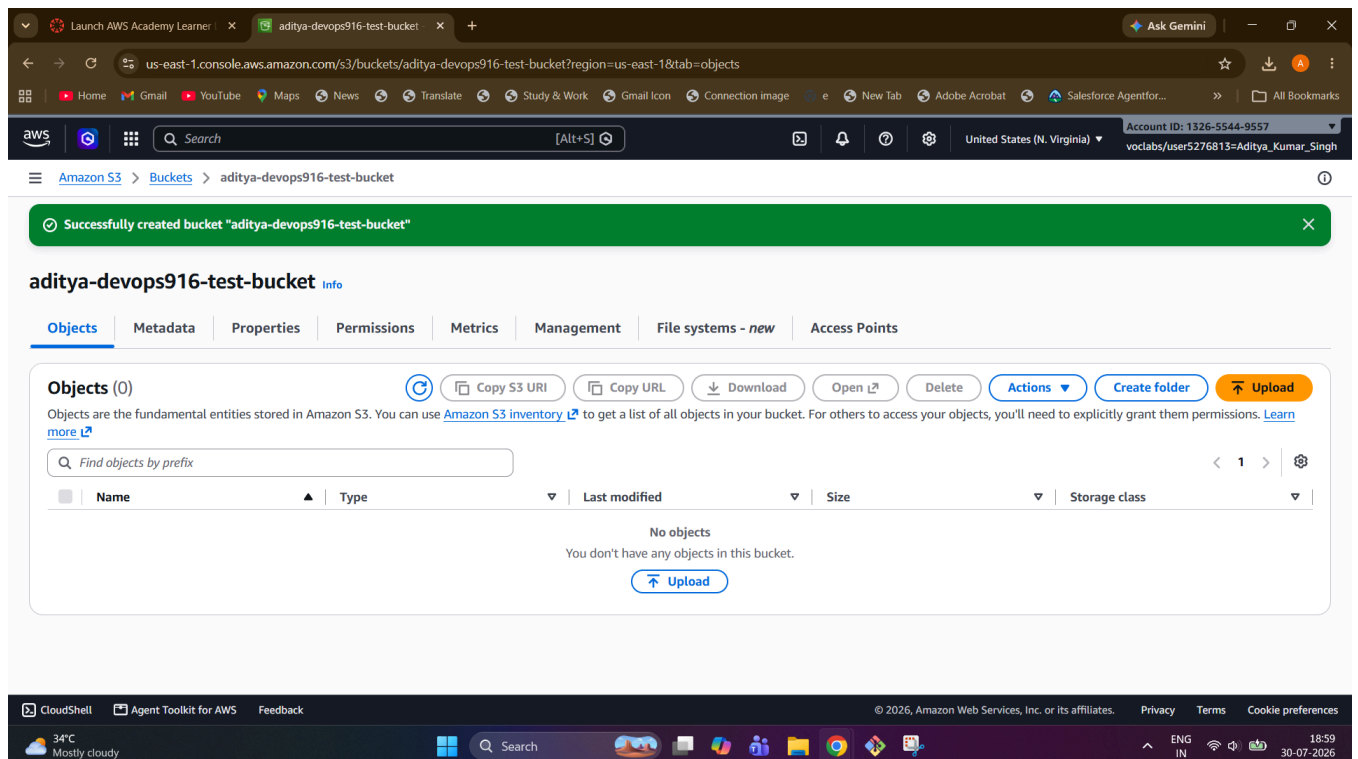
Amazon Simple Storage Service (S3) is a highly scalable, durable object storage service used to store and retrieve any amount of data from anywhere on the web. It supports fine-grained access control, versioning, and automated data lifecycle management, making it a foundational service for cloud storage. This experiment demonstrates creating an S3 bucket, uploading and managing files with versioning and lifecycle policies, and retrieving data using the AWS CLI.

Procedure

1. Create S3 Bucket

- Created a uniquely named S3 bucket using default settings.
- Ensured public access was blocked to keep the bucket secure by default.

s3BucketCreation.mp4



2. Upload Files

- Uploaded a test file (my-first-file.txt) to the bucket with default permissions.

s3FileUpload.mp4

Upload succeeded
For more information, see the [Files and folders](#) table.
After you navigate away from this page, the following information is no longer available.

Summary

Destination s3://aditya-devops916-test-bucket	Succeeded 1 file, 96.1 KB (100.00%)	Failed 0 files, 0 B (0%)
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Files and folders | Configuration

Files and folders (1 total, 96.1 KB)

Find by name

Name	Folder	Type	Size	Status	Error
2340.png	-	image/png	96.1 KB	Succeeded	-

3. Manage Files

- Enabled versioning on the bucket to preserve, retrieve, and restore every version of every object.
- Created a lifecycle rule to transition objects to S3 Glacier Flexible Retrieval after 30 days.

s3FileUpload.mp4

Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions
All storage classes 128K

Lifecycle rules (1)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Find lifecycle rules by name

Lifecycle rule name	Status	Scope	Current version acti...	Noncurrent versions...	Expired object delet...	Incomplete multipa...
move-to-glacier	Enabled	Entire bucket	Transition to Standard-IA	-	-	-

aditya-devops916-test-bucket

Objects | Metadata | **Properties** | Permissions | Metrics | Management | File systems - new | Access Points

Bucket overview

AWS Region US East (N. Virginia) us-east-1	Amazon Resource Name (ARN) arn:aws:s3::aditya-devops916-test-bucket	Creation date July 30, 2026, 18:56:05 (UTC+05:30)
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Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
Enabled

Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. [Learn more](#)

Disabled

4. Retrieve Files

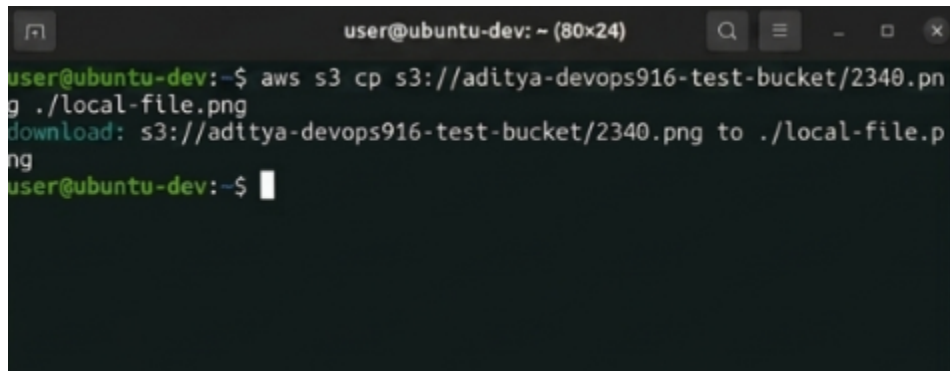
- Installed and configured the AWS CLI using:

```
aws configure
```

- Downloaded the file from the bucket using:

```
aws s3 cp s3://<bucket-name>/my-first-file.txt ./downloaded-file.txt
```

- Verified the downloaded file's content in the terminal.



```
user@ubuntu-dev: ~ (80x24)
user@ubuntu-dev:~$ aws s3 cp s3://aditya-devops916-test-bucket/2340.png ./local-file.png
download: s3://aditya-devops916-test-bucket/2340.png to ./local-file.png
user@ubuntu-dev:~$
```